

Macro-Financial Scenario Analysis

Methodology

Morningstar Quantitative Research

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Author

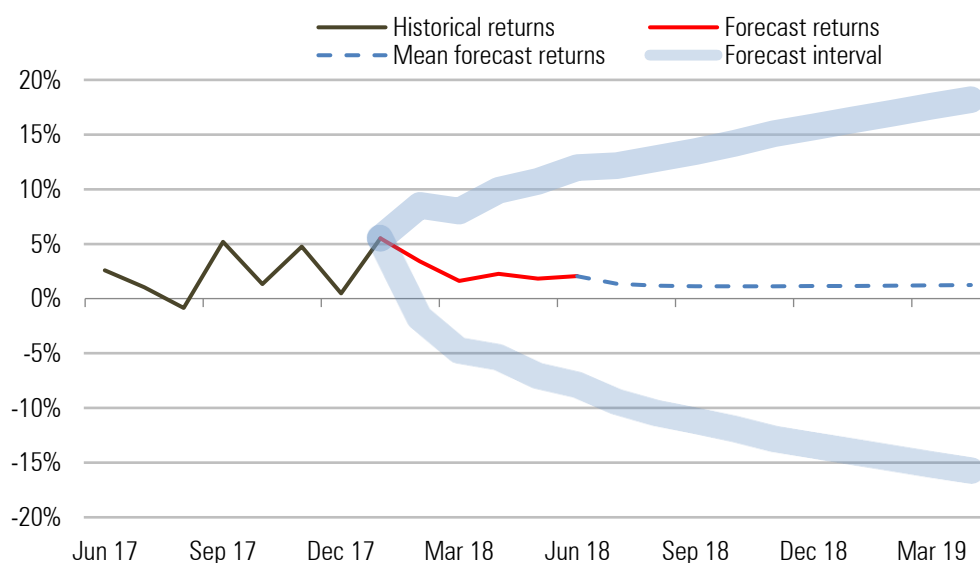
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Overview

Morningstar's Macro-Financial Scenario Analysis tool calculates the impact of user-specified macroeconomic and financial system shocks on forecast factor exposures and volatilities. With this tool the user can specify multiple shocks at different points in time and calculate the subsequent expected risk and return of the portfolio.

As of February 2018, the U.S. economy is approaching full employment. The Department of Labor reported in the January Jobs Report that wages are starting to grow. Under these conditions, the prospect of rising inflation looms. What will happen if the Consumer Price Index rises by 3% over the next five months?

Exhibit 1 Forecast Returns for Vanguard Small Cap ETF Where CPI Rises 3% Over 5 Months



Source: Morningstar, Inc. Data as of Feb. 6, 2018.

Exhibit 1 plots the historical returns for Vanguard Small Cap ETF VB along with the forecast returns under the condition that CPI increases by 3% over the next five months. Exhibit 2 shows the specific CPI scale factors used in this scenario. In other words, the value of CPI at the end of February is 1.01 times the last reported value of CPI at the end of January. The dotted blue line in Exhibit 1 represents the mean forecast six months from now and beyond. As expected, the mean forecast returns to VB's

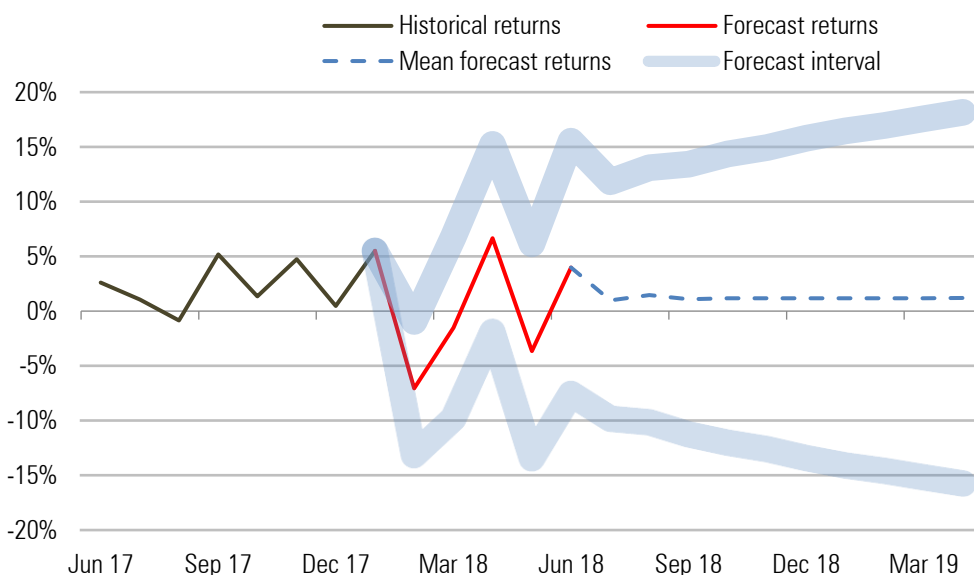
historical mean, which is approximately 5% below its value at the end of January 2018. The thick blue lines in Exhibit 1 represent the volatility forecast interval.

Exhibit 2 Macro-Financial Shock Input for Vanguard Small Cap ETF Scenarios

Date	CPI	S&P 500	GDP
	Last Value Scale		
02/28/2018	1.01	0.96	1
03/31/2018	1.015	0.92	1.005
04/30/2018	1.02	0.94	1.01
05/31/2018	1.025	0.93	1.015
06/30/2018	1.03	0.95	1.02

What if, in addition to CPI rising by 3% over the next five months, the S&P 500 loses 8% during the first two months and eventually recovers to a 5% loss during the same period, as summarized in the S&P 500 column of Exhibit 2? The impact of such a scenario on VB is plotted in Exhibit 3.

Exhibit 3 Forecast Returns for Vanguard Small Cap ETF Where CPI Rises 3% and S&P 500 Falls 5% Over 5 Months



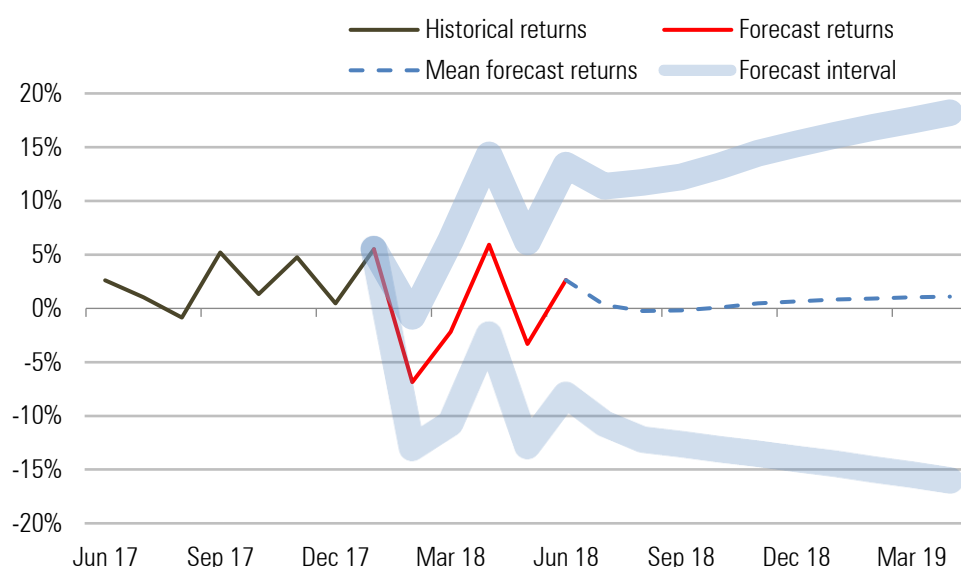
Source: Morningstar, Inc. Data as of Feb. 6, 2018.

Finally, what would happen under the same conditions (CPI rises 3%, S&P 500 loses 5% over the same period) if the International Monetary Fund's GDP projections summarized in the GDP column of Exhibit 2 are also included in the analysis? Exhibit 4 demonstrates that the projected GDP growth would slightly dampen the volatility of the scenario shown in Exhibit 3.

All three scenarios are analyzed using Morningstar's Macro-Financial Scenario Analysis tool. By adding new shocks to a given scenario, the user is in effect selecting the correlation among the constrained

macro-financial variables and evaluating the impact on forecast factor exposures and volatilities. This gives the user the flexibility to explore the implications of multifaceted scenarios in an intuitive way.

Exhibit 4 Forecast Returns for Vanguard Small Cap ETF Where CPI Rises 3%, S&P 500 Falls 5%, and GDP Grows 2%



Source: Morningstar, Inc. Data as of Feb. 6, 2018.

Macro-Financial Variables

Four predefined lists of macro-financial variables are provided as defaults to model the Asia-Pacific, eurozone, U.K., and U.S. economies, respectively. The user can select additional economic variables to include in the model, or to create a model from scratch. As most economic variables are reported monthly, all other variables (that is, daily, weekly, or quarterly variables) included in the macro-financial model are resampled monthly to ensure that all variables have the same frequency.

Statistical forecasting methods typically require that the time series are made approximately stationary, which implies that the mean and variance of the time series are constant over time. Financial data tends to have strong trends and seasonality; it is not, as a rule, stationary. The simplest way to make a time series stationary is to compute the difference between consecutive observations. Moreover, logarithms can help stabilize the variance of a time series. Therefore, the log-difference of the macro-financial data is calculated and standardized to determine the z-score. If there are any negative values in the variables, the entire time series is shifted by the absolute value of the minimum number to calculate the log-difference.

Macro-Financial Shocks

The user can select the macro-financial variables to be shocked and the duration of the shock in months. Subsequent shocks can be specified by the end user as well. For example, the user can select a 10% increase in Real GDP Percent Change in month one followed by a 5% increase in month two, as shown in Exhibit 5. The user can then specify a 10% increase in Net Exports in the first month and a 20%

increase in the second month. The user can also specify the shock level based on standard deviation (for example, 0.5σ Net Exports on Feb. 28, 2018) or on the actual value of the variable (for example, 4.2% Real GDP on Feb. 28, 2018).

Exhibit 5 Macro-Financial Shock Input

Date	Real GDP	Net Exports	Real GDP	Net Exports	Real GDP	Net Exports
	Last Value Scale		Standard Deviation Scale		Actual Value	
02/28/2018	1.1	1.1	0.5	0.5	4.2%	-\$500B
03/31/2018	1.05	1.2	1.0	1.0	4.5%	-\$510B

Principal Component Analysis

Principal Component Analysis, or PCA, is performed on the log-differenced data to reduce the dimensionality of the analysis and to discern the driving macro-financial factors of the model. The number of PCA components is determined by the minimum mean squared error between the time series of the primary variable of interest, which is simply the first variable of the first shock selected by the end user, and the time series reconstructed from the PCA mixing matrix of the primary variable of interest. The maximum number of PCA components is limited to half the total number of variables in the analysis to focus on the driving macro-financial factors.

Risk Factor Premia Linkage Model

To find a stock's true sources of returns, Morningstar provides several risk models constructed for the purposes of assessing risk in different regions and asset classes. Each model has a unique collection of factor returns that indicate the returns per units of factor risk. Depending on the choice of risk model, the appropriate risk factor premia, or RFP, are resampled using the monthly cumulative product and the PCA components are aligned with the most recent RFP data. Multivariate linear regression analysis of the PCA components is performed on the RFP data to create an ordinary least-squares model, which links the macro-financial shock to the underlying revenue drivers of returns.

Vector Autoregressive Processes

An autoregressive process for a univariate time series y_t takes the following form:

$$y_t = v + \alpha_1 y_{t-1} + \dots + \alpha_p y_{t-p} + u_t,$$

where y_t , $\alpha_1 y_{t-1} \dots \alpha_p y_{t-p}$, and u_t are random variables. When multiple time series are considered, the notation in the equation above is extended to obtain a vector autoregressive, or VAR, process:

$$\mathbf{y} = \mathbf{v} + \sum_{i=1}^p \mathbf{A}_i \mathbf{y}_{t-i} + \mathbf{u},$$

where $\mathbf{v}$ is a fixed K -dimensional vector of intercept terms, $\mathbf{u}$ is a K -dimensional white noise or innovation process with a covariance matrix $\Sigma_{\mathbf{u}} = E(\mathbf{u}_t \mathbf{u}_t^T) = \text{Cov}(\mathbf{u}_t)$, and A_i are $K \times K$ coefficient matrices:

$$A_i = \begin{bmatrix} \alpha_{11} & \cdots & \alpha_{1K} \\ \vdots & \ddots & \vdots \\ \alpha_{K1} & \cdots & \alpha_{KK} \end{bmatrix}.$$

A VAR model is created with the PCA components calculated in the step outlined above to determine the white noise covariance matrix $\Sigma_{\mathbf{u}}$, which is used in solving an optimization problem.

Optimization Problem

The standard form of an optimization problem is to maximize an objective function subject to constraints.

By maximizing the log likelihood of the multivariate normal probability density function, or pdf,

$2\pi^{-\frac{K}{2}} |\Sigma|^{-\frac{1}{2}} e^{-\frac{1}{2}(\mathbf{u}-\boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{u}-\boldsymbol{\mu})}$, where $\boldsymbol{\mu} = \mathbf{0}$, the objective function of the optimization problem is defined as

$$f(\mathbf{u}_t) = -\mathbf{u}_t^T \Sigma^{-1} \mathbf{u}_t.$$

Recalling that PCA decomposition is a function of

$$\mathbf{X}_m = \underbrace{\mathbf{PCS}}_{t \times nc} \times \underbrace{\mathbf{M}}_{nc \times x},$$

where x represents the individual macro-financial variables, nc represents the number of PCA components, $\mathbf{PCS}$ are the underlying PCA components, and $\mathbf{M}$ is the mixing matrix tying the two models together, the shock constraint can be written in terms of the VAR model. Specifically, by multiplying both sides of the VAR equation by the column of mixing matrix $\mathbf{M}$ associated with the shock variable x_s , a constraint equation is found in terms of the shock term

$$M_{x_s}^T y_t = M_{x_s}^T \sum_{i=1}^p A_i y_{t-i} + M_{x_s}^T u_t,$$

where $M_{x_s}^T y_t$ is the shock value.

The optimization problem ($\nabla f + \lambda \nabla g = \mathbf{0}$) is solved using Lagrange multipliers, where $f(\mathbf{u}_t) = -\mathbf{u}_t^T \Sigma^{-1} \mathbf{u}_t$ subject to the constraint equation. Therefore,

$$F(\mathbf{u}_t, \lambda) = -\mathbf{u}_t^T \Sigma^{-1} \mathbf{u}_t + \lambda \left(M_{x_s}^T \sum_{i=1}^p A_i y_{t-i} + M_{x_s}^T u_t - M_{x_s}^T y_t \right).$$

Setting the partial derivatives to zero, the following set of equations is obtained:

$$\begin{array}{ccccccc} -2\Sigma_{11}^{-1}u_{t_1} & + \cdots + & -u_{t_{nc}}\Sigma_{nc1}^{-1} & + \lambda M_{x_{s1}} & = & 0 \\ \vdots & -2\Sigma_{1nc}^{-1} & \vdots & + \vdots & = & 0 \\ -u_{t_1}\Sigma_{1nc}^{-1} & + \cdots + & -2\Sigma_{ncnc}^{-1}u_{t_{nc}} & + \lambda M_{x_{snc}} & = & 0 \\ M_{x_{s1}}u_{t_1} & + \cdots + & M_{x_{snc}}u_{t_{nc}} & + 0 & = & M_{x_s}^T y_t - M_{x_s}^T A_p y_{t-p} \end{array}$$

which is a linear equation in u_{t_1} and λ , equivalent to a matrix equation of the form $\mathbf{Ax} = \mathbf{b}$, where

$$\mathbf{A} = \begin{bmatrix} -2\boldsymbol{\Sigma}_{11}^{-1} & \cdots & -\boldsymbol{\Sigma}_{nc1}^{-1} & M_{x_{s1}} \\ \vdots & -2\ddots & \vdots & \vdots \\ -\boldsymbol{\Sigma}_{1nc}^{-1} & \cdots & -2\boldsymbol{\Sigma}_{ncnc}^{-1} & M_{x_{snc}} \\ M_{x_{s1}} & \cdots & M_{x_{snc}} & + 0 \end{bmatrix}$$

and

$$\mathbf{b} = \begin{bmatrix} 0 \\ \vdots \\ 0 \\ M_{x_s}^T y_t - M_{x_s}^T A_p y_{t-p} \end{bmatrix}.$$

As soon as the first shock duration is complete, the second shock begins, and the optimization problem is solved on a time-step basis.

Vector Autoregressive Forecast

Once the impact of the shocks is determined through the solution to the optimization problem outlined above, a second VAR model is created with the underlying PCA components, which now include new time steps incorporating the effect of the shocks. The minimum mean squared error, or MSE, prediction for a forecast horizon h is calculated as the conditional expected value $E(y_{t+h})$. Specifically, the prediction minimizes the MSE of each component of y_t .

Risk Factor Premia Forecast

With the linkage model described above, the risk factor premia forecast is calculated. The variance during the shock is determined by computing the Euclidean distance between a rolling window of past risk factor premia and the forecast risk factor premia, and then assembling a distribution weighted by that metric. As the variance of the errors in a random walk model grow linearly with time, the standard deviation of the time step ahead forecast error is proportional to the square root of time for the risk factor premia forecast.

Total Return Forecast

With the risk factor premia forecast, the total return of a given portfolio is calculated based on the last available factor exposures:

$$r_{t+m} = X_t f_{t+m},$$

where r_{t+m} is the return of a portfolio, t is the most recent month end, m represents the month of the scenario analysis forecast, X_t is a 1×36 vector of factor exposures at time t , and f_{t+m} is a vector of forecast risk factor premia at time $t + m$.

Appendix A: Model Variables

The macro-financial variables used in all the example scenarios are shown in Exhibit 6.

Exhibit 6 Forecast Macro-Financial Variables Available

Description	Frequency
Consumer Price Index for All Urban Consumers: Rent of Primary Residence	M
Industrial Production Index	M
Nonfinancial Corporate Business; Corporate Equities; Liability, Level	Q
Nonfinancial Corporate Business; Credit Market Debt as a Percentage of Net Worth	Q
Consumer Price Index for All Urban Consumers: All Items	M
Nonfinancial Corporate Business: Profits Before Tax (Without IVA and CCAdj)	Q
Nonfarm Business Sector: Real Compensation Per Hour	Q
Government Consumption Expenditures and Gross Investment	Q
Real Government Consumption Expenditures and Gross Investment	Q
Gross Domestic Product	Q
Real Gross Domestic Product	Q
Gross Domestic Product: Implicit Price Deflator	Q
Gross Private Domestic Investment	Q
Net Exports of Goods and Services	Q
Nonfinancial Corporate Business: Profits After Tax (without IVA and CCAdj)	Q
Nonfarm Business Sector: Real Output Per Hour of All Persons	Q
Private Residential Fixed Investment	Q
Unemployment Rate: 25 to 54 Years, Men	M
Civilian Unemployment Rate	M
Total Population: All Ages Including Armed Forces Overseas	M
Consumer Price Index for All Urban Consumers: All Items Less Food and Energy	M
Housing Starts: Total: New Privately Owned Housing Units Started	M
Personal Consumption Expenditures	M
Government Total Expenditures	Q
Monthly Supply of Houses in the United States	M
Index of Aggregate Weekly Hours: Production and Nonsupervisory Employees: Total Private Industries	M

Exhibit 5 Forecast Macro-Financial Variables Available

Description	Frequency
Capacity Utilization: Total Industry	M
4-Week Moving Average of Initial Claims	W
Initial Claims	W
All-Transactions House Price Index for the United States	Q
Crude Oil Prices: West Texas Intermediate (WTI) -- Cushing, Oklahoma	M
Total Business Inventories	M
Retail and Food Services Sales	M
Total Business Sales	M
Manufacturers' New Orders: Durable Goods	M
Manufacturers' New Orders: Nondefense Capital Goods Excluding Aircraft	M
Total Unemployed, plus all marginally attached workers plus total employed part time for economic reasons	M
Job Openings: Total Nonfarm	M
S&P 500	D
10-Year Treasury Constant Maturity Rate	D

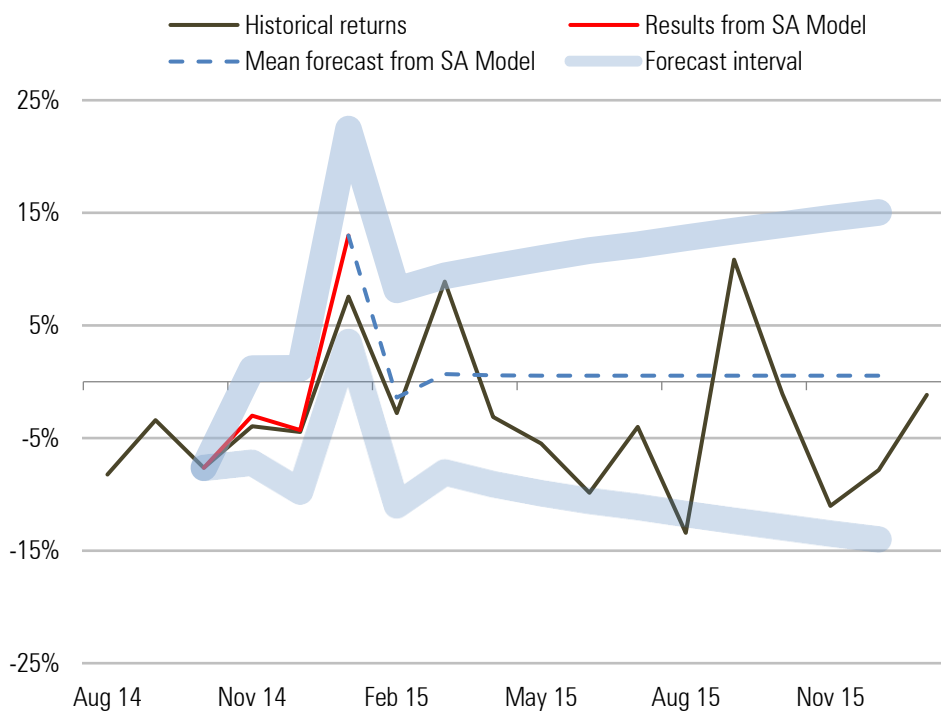
Source: Morningstar, Inc.

Appendix B: Historical Backtest

Oil Shock of 2014

At the end of 2014, oil prices fell from \$93.21 per barrel in September to \$47.22 per barrel in January 2015. For this historical example, the oil price is shocked for two months from November through December. As the oil price recovered slightly in January 2015, a subsequent shock is provided to capture the effect of that change. Exhibit 3 shows the historical returns for Energy Select Sector SPDR ETF XLE along with the scenario analysis mean forecast and volatility forecast interval. The list of macro-financial variables used in this example scenario is shown in Appendix A.

Exhibit 6 Forecast Returns for Energy Select Sector SPDR ETF



Source: Morningstar, Inc. Data as of Feb. 6, 2018.

About Morningstar® Quantitative Research

Morningstar Quantitative Research is dedicated to developing innovative statistical models and data points, including the Morningstar Quantitative Rating, the Quantitative Equity Ratings and the Global Risk Model.

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